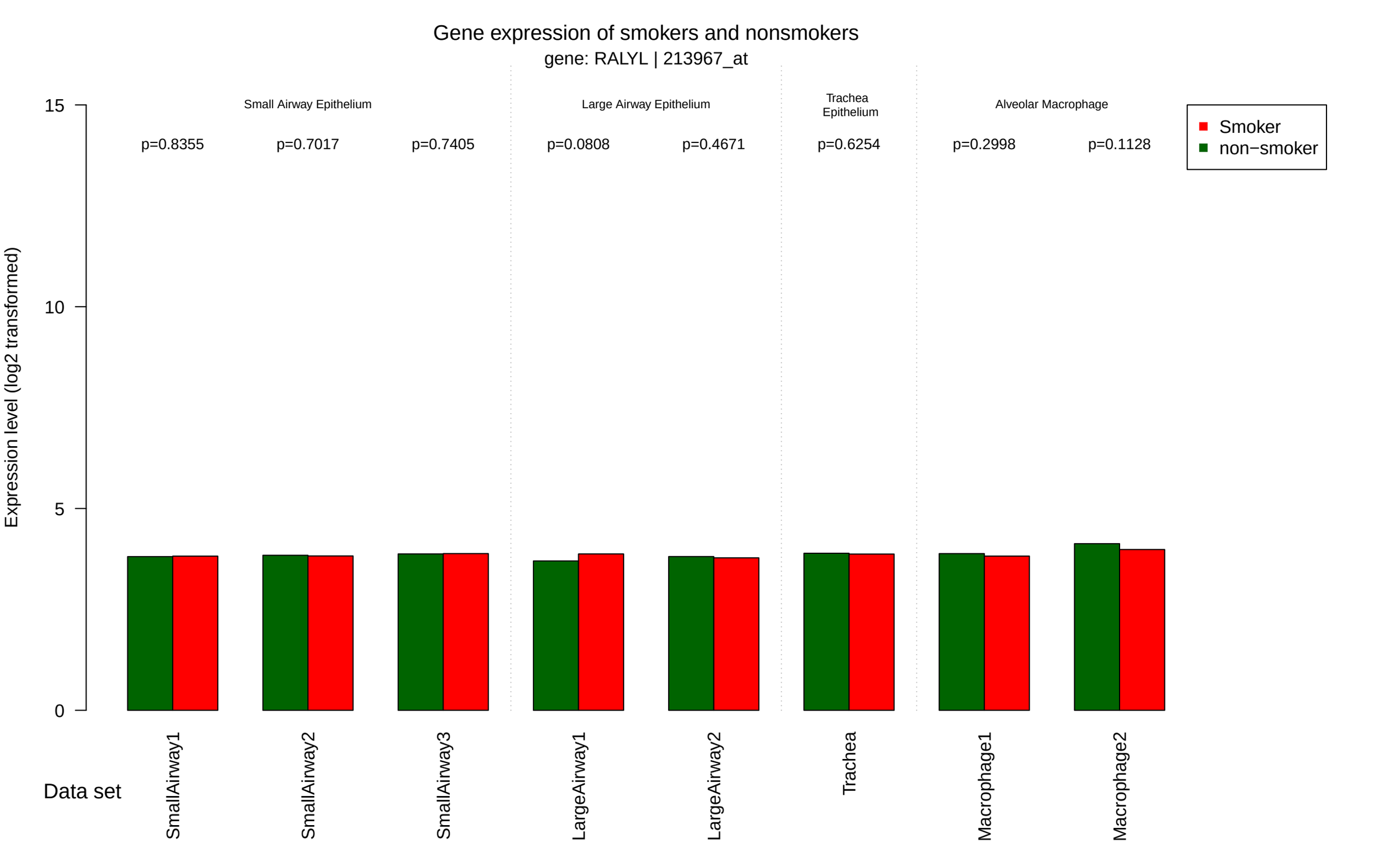
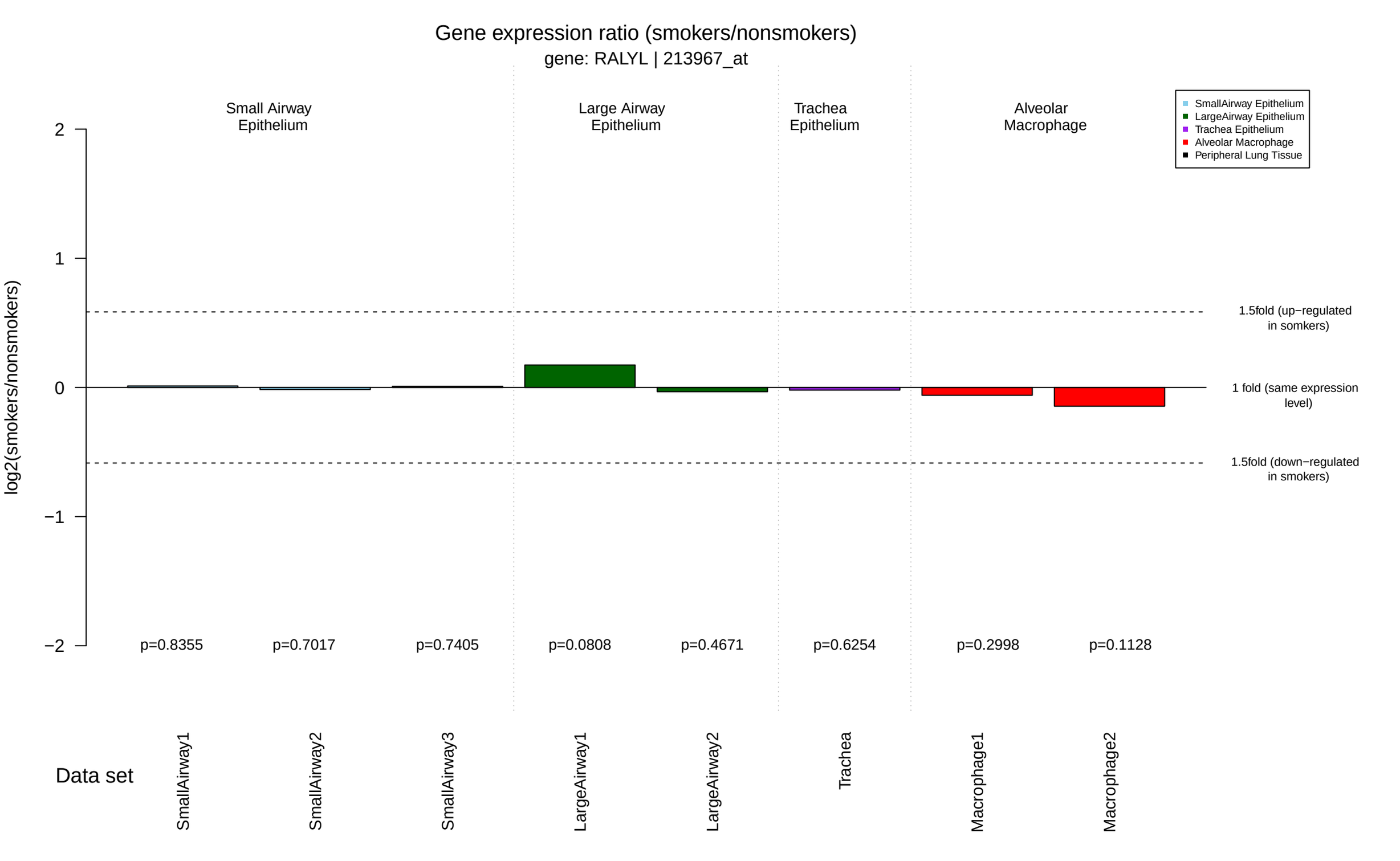
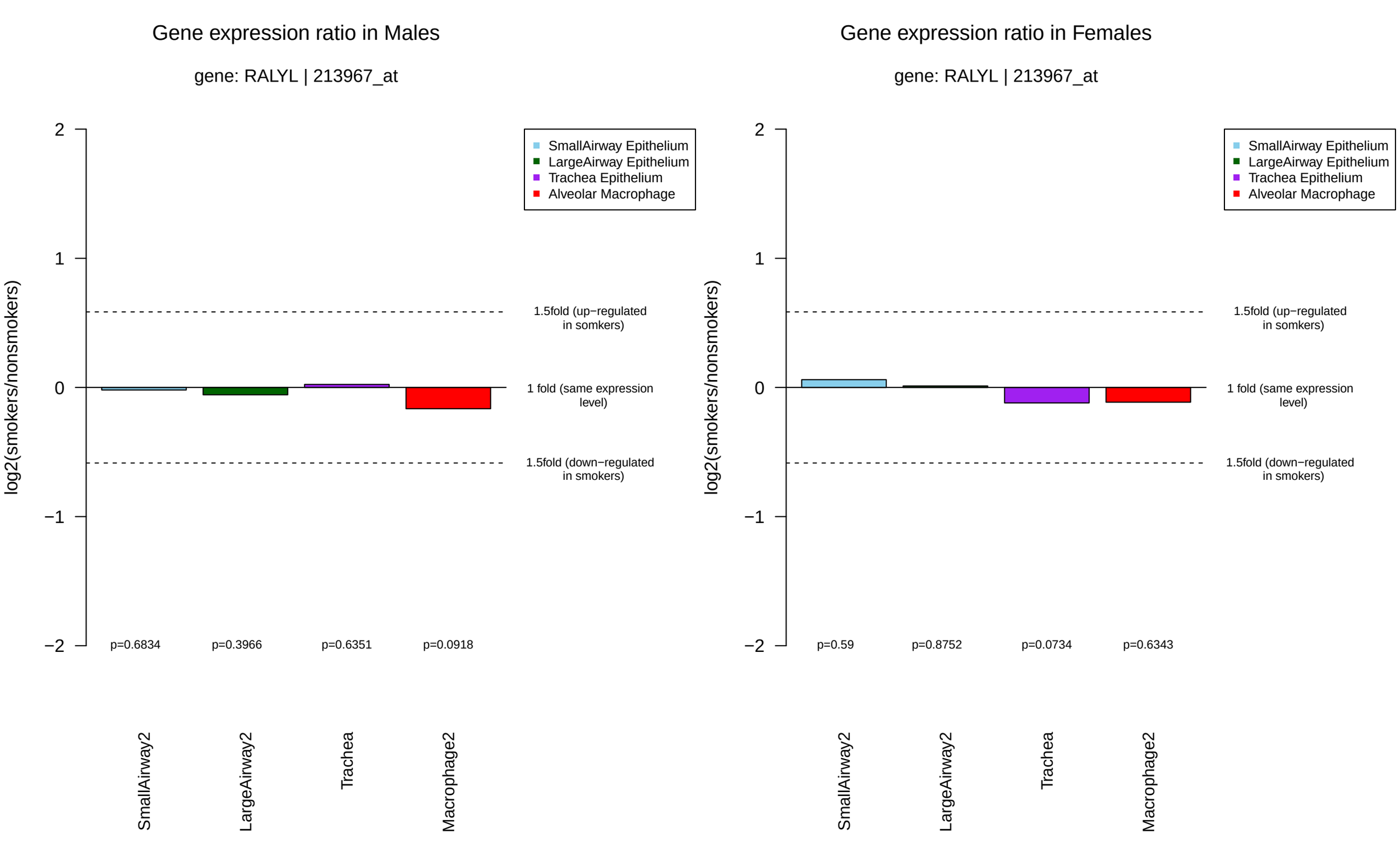
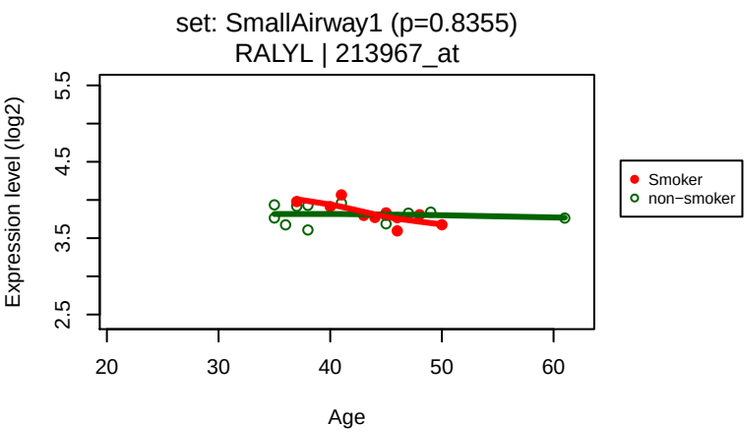
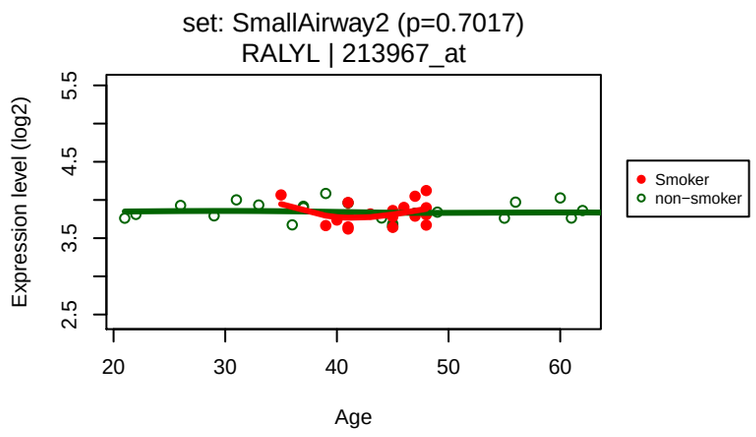
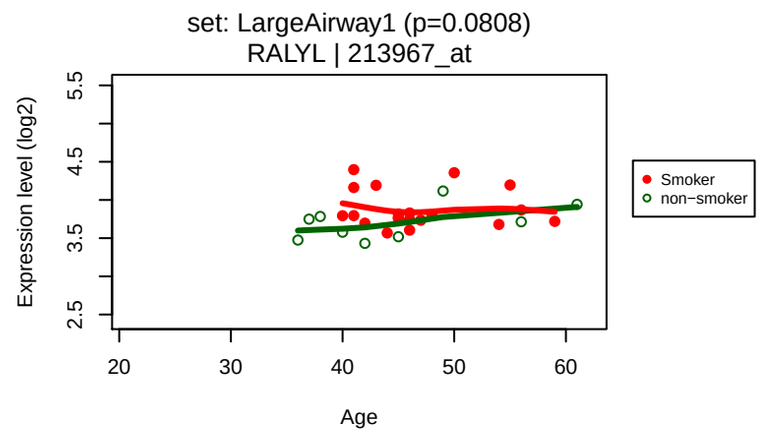

Gene expression of smokers and nonsmokers
gene: RALYL | 213967_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8355
p=0.7017
p=0.7405
p=0.0808
p=0.4671
p=0.6254
p=0.2998
p=0.1128
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

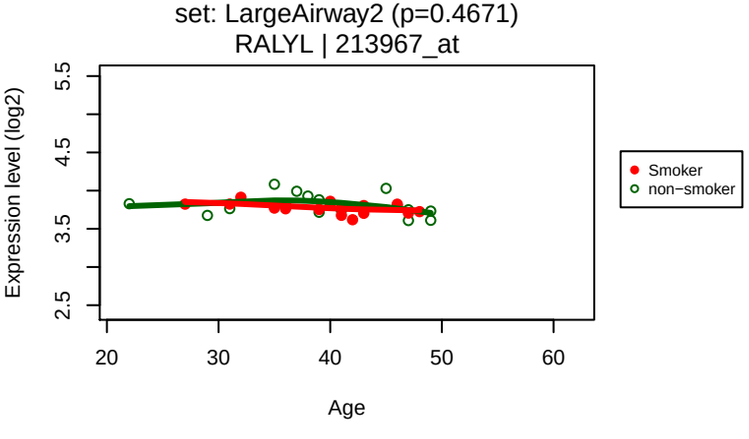

Gene expression ratio (smokers/nonsmokers)
gene: RALYL | 213967_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8355
p=0.7017
p=0.7405
p=0.0808
p=0.4671
p=0.6254
p=0.2998
p=0.1128
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

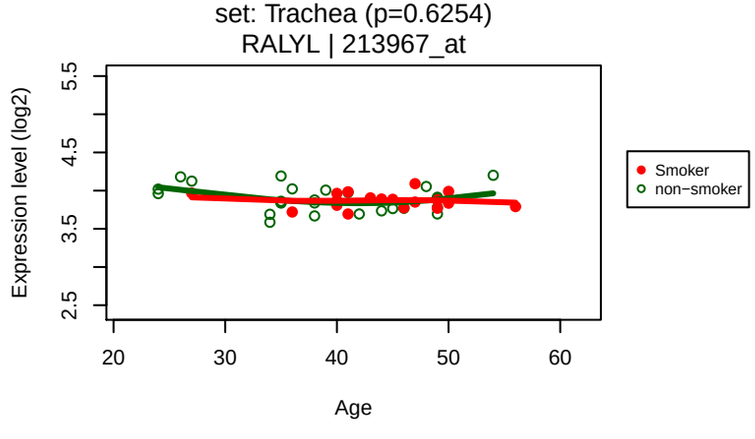

Gene expression ratio in Males
gene: RALYL | 213967_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6834
p=0.3966
p=0.6351
p=0.0918
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALYL | 213967_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.59
p=0.8752
p=0.0734
p=0.6343
SmallAirway2
LargeAirway2
Trachea
Macrophage2

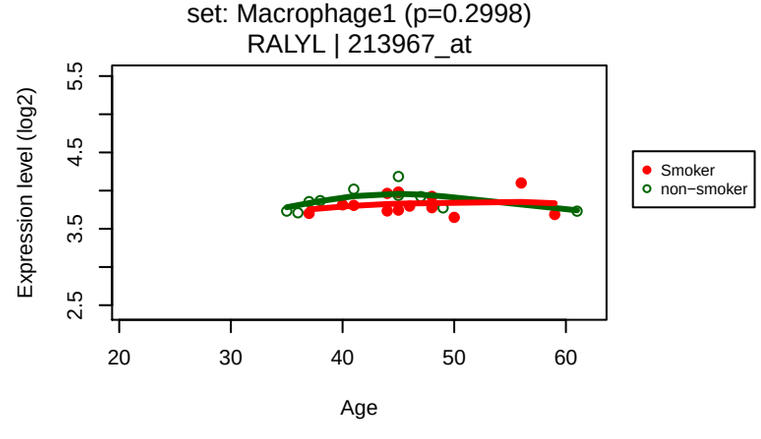

set: SmallAirway1 (p=0.8355)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker

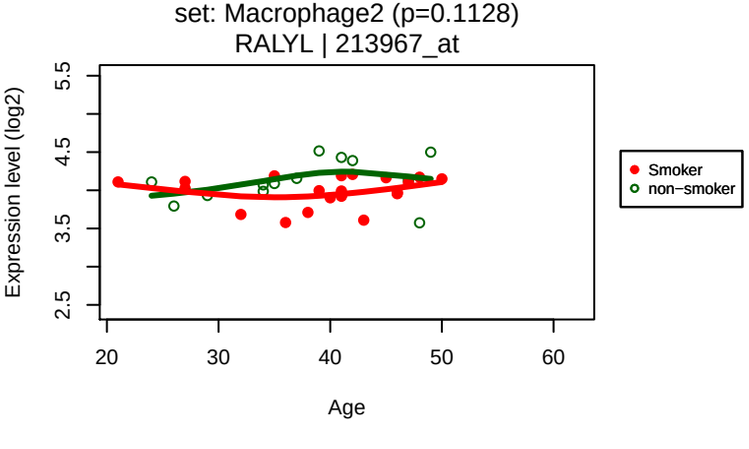

set: SmallAirway2 (p=0.7017)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker

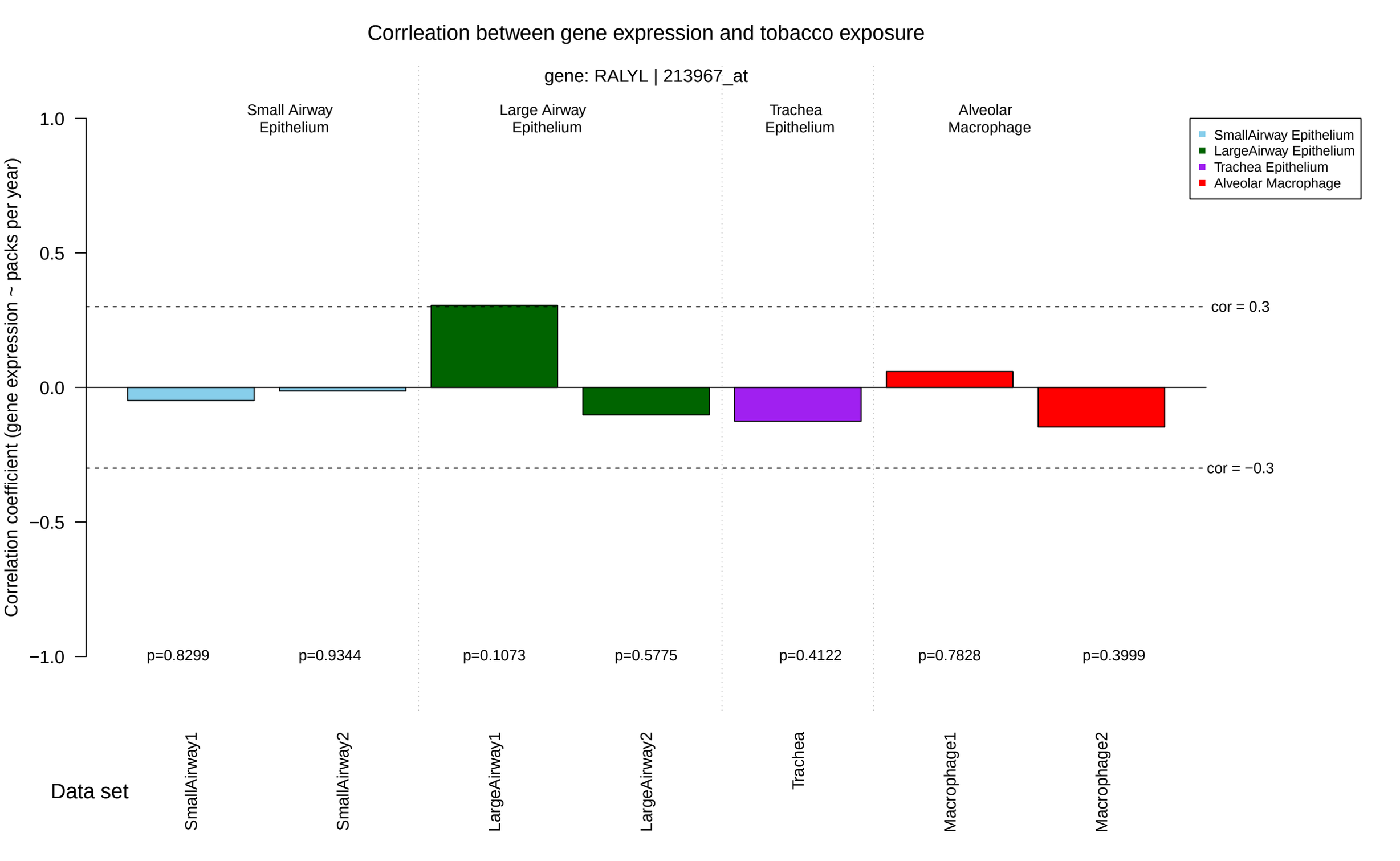

set: LargeAirway1 (p=0.0808)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4671)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6254)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2998)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1128)
RALYL | 213967_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALYL | 213967_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8299
p=0.9344
p=0.1073
p=0.5775
p=0.4122
p=0.7828
p=0.3999
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2